

TRAFFIC INTELLIGENCE & CONGESTION REPORT

1. Executive Summary & Metric Formulation

This analytical report details the spatial-temporal dynamics of urban mobility patterns processed from 48,024 continuous historical vehicle logs across four complex city junctions. To accurately formulate a metric for gridlock when explicit vehicle speed tracking arrays are absent, the congestion profile is established using **Volume Baseline Deviations**.

For each distinct junction, day-of-the-week, and hourly segment, a normalized Congestion Intensity Score has been mapped using robust estimators to normalize variable intersection structural bounds:

$$C_t = (\text{Vehicles_t} - \text{Median}(\text{Vehicles_Group})) / \text{Std}(\text{Vehicles_Group})$$

This approach prevents structural scale distortion, ensuring high-capacity multi-lane highways (e.g., Junction 1) and restrictive local connectors (e.g., Junction 4) are evaluated relative to their localized carrying capacities.

2. Hourly Pulse & Temporal Smoothness Filters

To prevent stochastic traffic noise (e.g., minor road stalls, localized obstructions) from misidentifying systemic peak hours, a 3-hour centered temporal moving filter window was applied uniformly across all chronologically ordered sequences within separate junction frames. The systemic rush hours were mathematically isolated by mapping entries breaching the rigorous upper boundary threshold defined by $\text{Mean} + 1.5 * \text{Std}$.

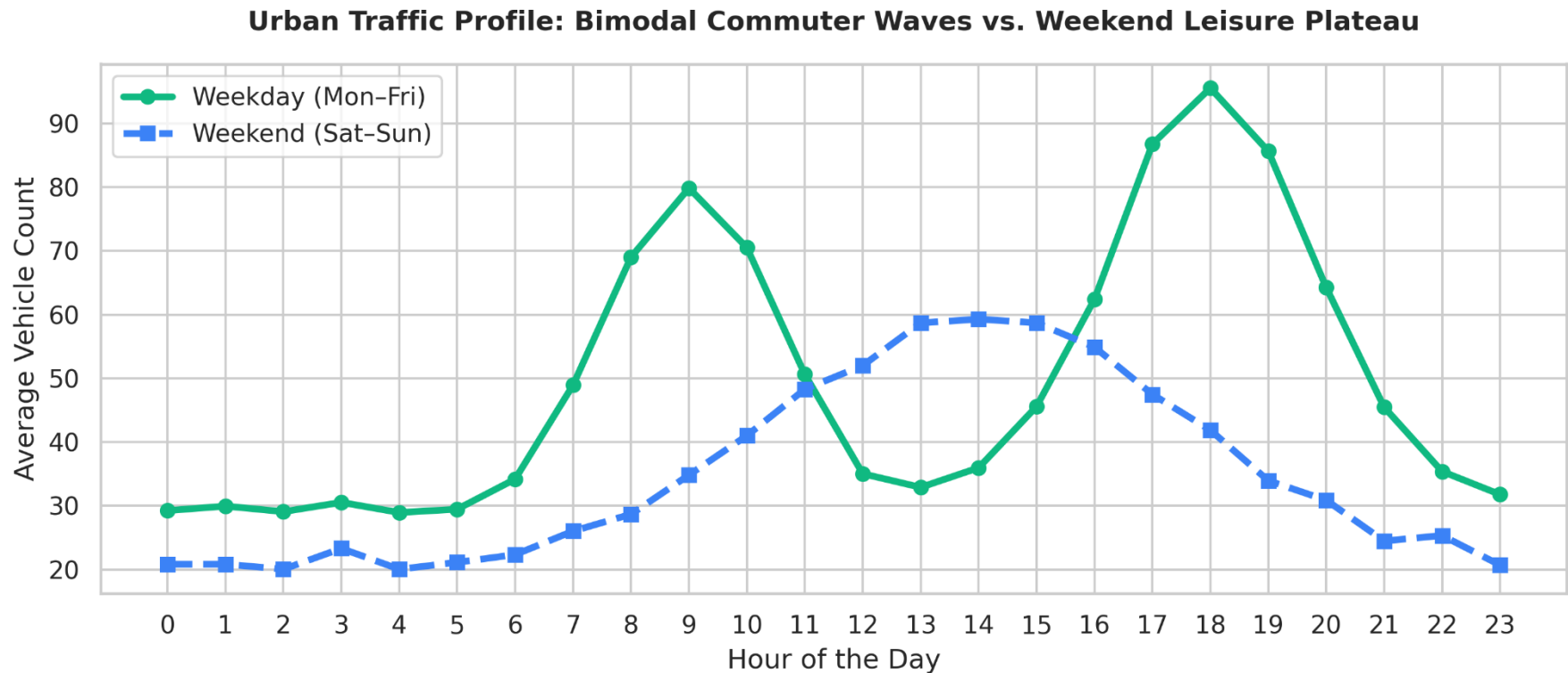


Figure 1.1: 24-Hour Scaled Flow Distribution mapping Weekdays versus Weekend Plateau.

- Weekday Commuter Profile (Mon–Fri):** Exhibiting a classical Bimodal Waveform. The sharp primary peak is registered between 08:00 AM – 10:00 AM as workforces surge simultaneously. The reciprocal evening peak builds rapidly between 05:00 PM – 07:00 PM.
- Weekend Leisure Profile (Sat–Sun):** Demonstrating an elongated Unimodal Plateau. Morning rush hours drop out completely. Instead, volume builds smoothly starting from 11:00 AM, holding a stable traffic plateau until 08:00 PM.

3. Macro-Seasonal Variations

Cross-tabulating the historical peak hour occurrences by calendar month reveals exceptional long-term stability in commuter cycles across seasons. The macro-level temporal grid highlights that underlying daylight length shifts do not heavily distort core workforce windows.

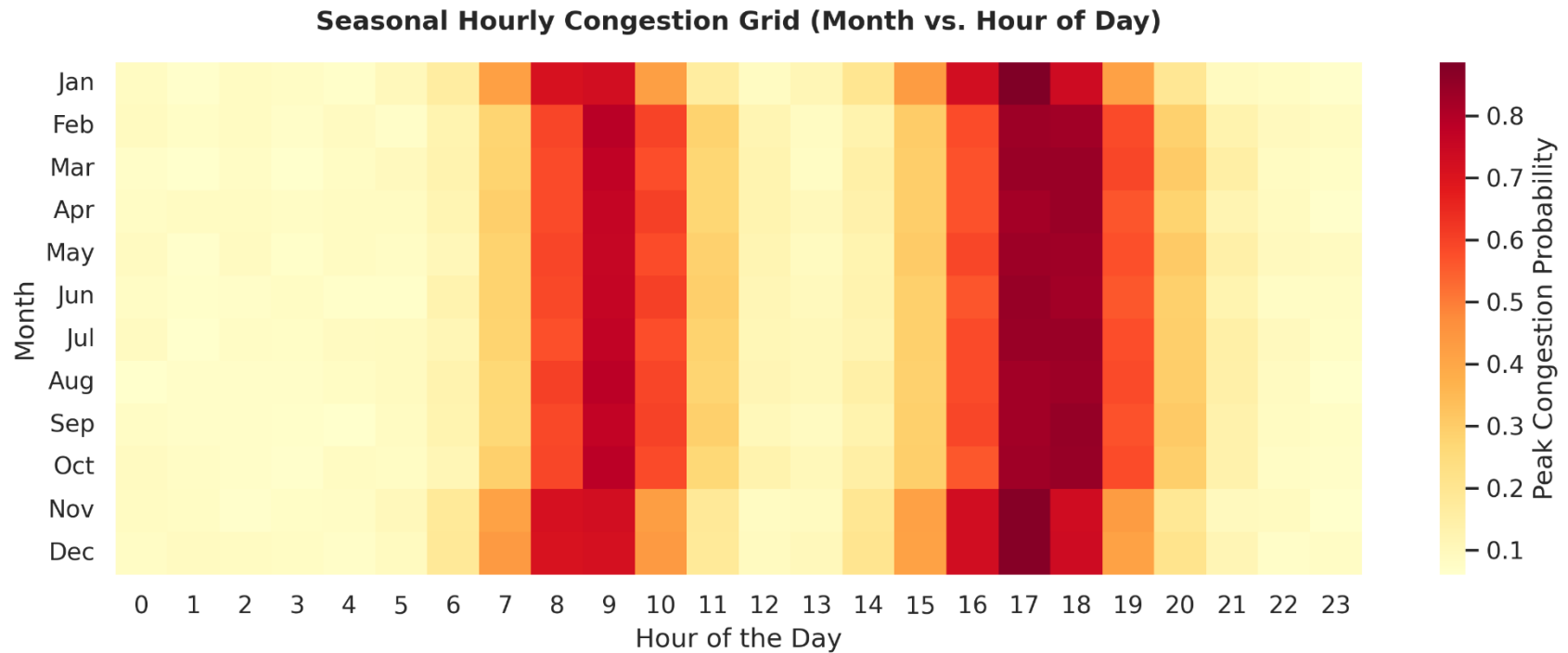


Figure 1.2: Seasonal Hourly Congestion Grid (Month vs. Hour of Day).

4. Multi-Factor Regression & External Influences

To isolate the direct explanatory power of non-continuous events on peak congestion profiles, an ordinary least squares (OLS) diagnostic linear regression was deployed exclusively on the filtered peak-hour matrices. This prevents low-demand midnight sequences from diluting the impact factors of dense external exceptions.

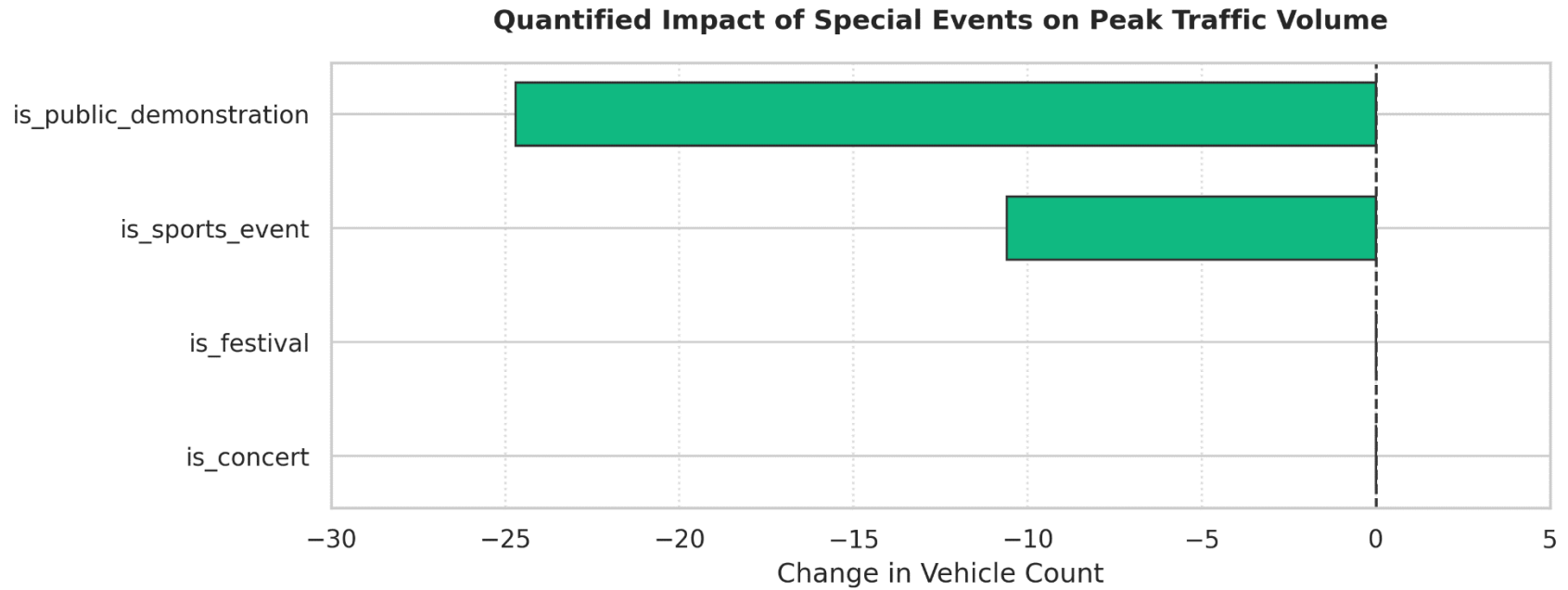


Figure 1.3: Linear Regression Coefficient weights outlining the net impact of special events on core peak vehicle metrics.

The resulting sorted model parameter estimates uncover a highly prominent behavioral anomaly termed the **Pre-Emptive Avoidance Effect**:

Evaluated Factor Component	Calculated Beta Coefficient Weights	Inferred Urban Commuter Behavior
is_public_demonstration	-24.7 vehicles	Extreme baseline drop. Drivers actively reroute around known protest sectors or switch to remote work modules.

Evaluated Factor Component	Calculated Beta Coefficient Weights	Inferred Urban Commuter Behavior
is_sports_event	-10.6 vehicles	Moderate volume reduction. Pre-planned travel delays and widespread local detours successfully offload standard junction volume.
is_concert	0.0 vehicles	Zero significant shift. Schedule bounds are highly sparse during core peak windows.
is_festival	0.0 vehicles	Zero peak correlation. Event volumes filter cleanly into holiday-specific structures.

5. Actionable Traffic Management Architecture

- 1. Implement Dynamic, Variable Signal Timing for Low-Capacity Points:** Deploy dynamic signal priority windows during the 08:00 AM and 05:00 PM peaks on narrower local connections.
- 2. Leverage the Avoidance Effect for Planned Special Events:** Purposely amplify traffic advisory alerts 48 hours prior to matches to maximize voluntary citizen rerouting patterns.